

Workshop

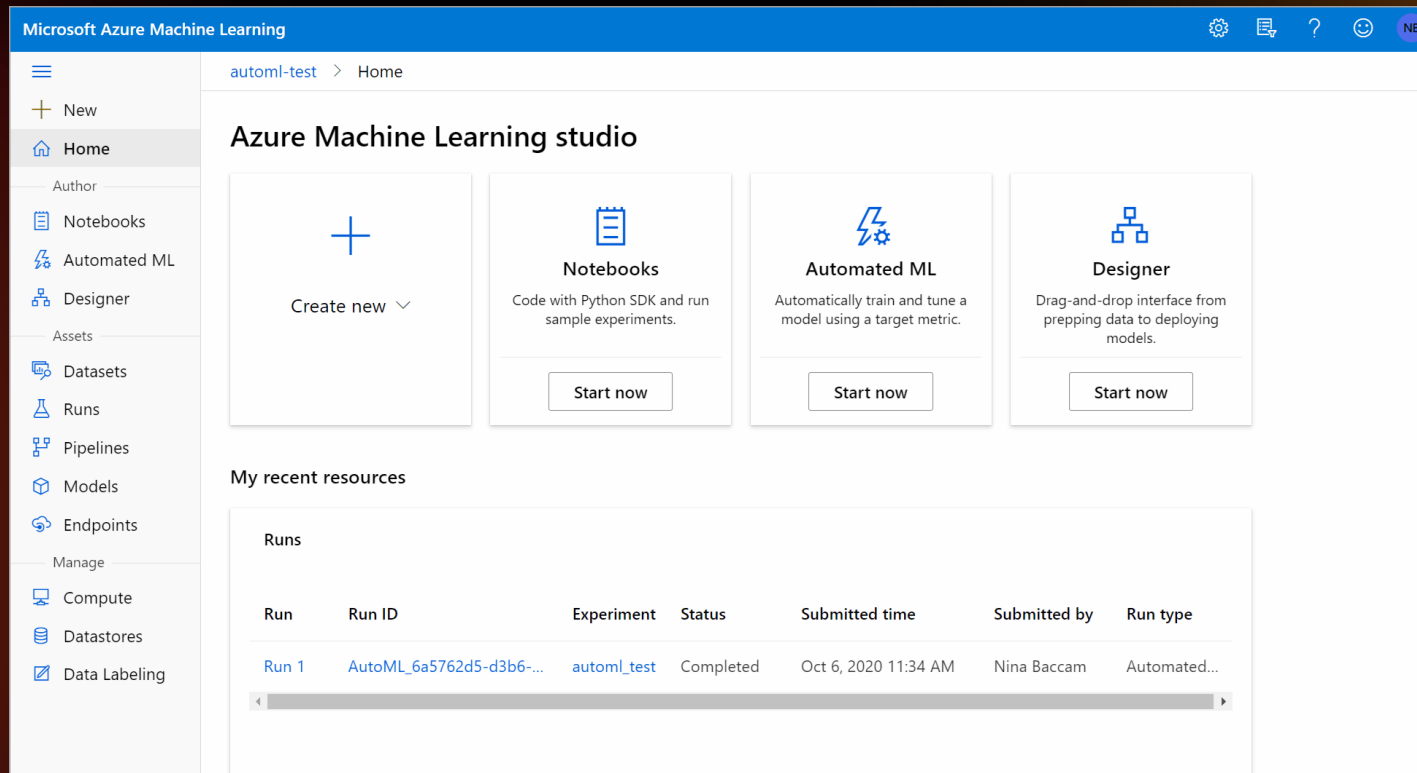
HAGI KGI

Explore the Azure Machine Learning workspace resources and assets

Introduction

Azure Machine Learning

Azure Machine Learning provides a platform for data scientists to train, deploy, and manage their machine learning models on the Microsoft Azure platform. It provides a comprehensive set of resources and assets to train and deploy effective machine learning models.

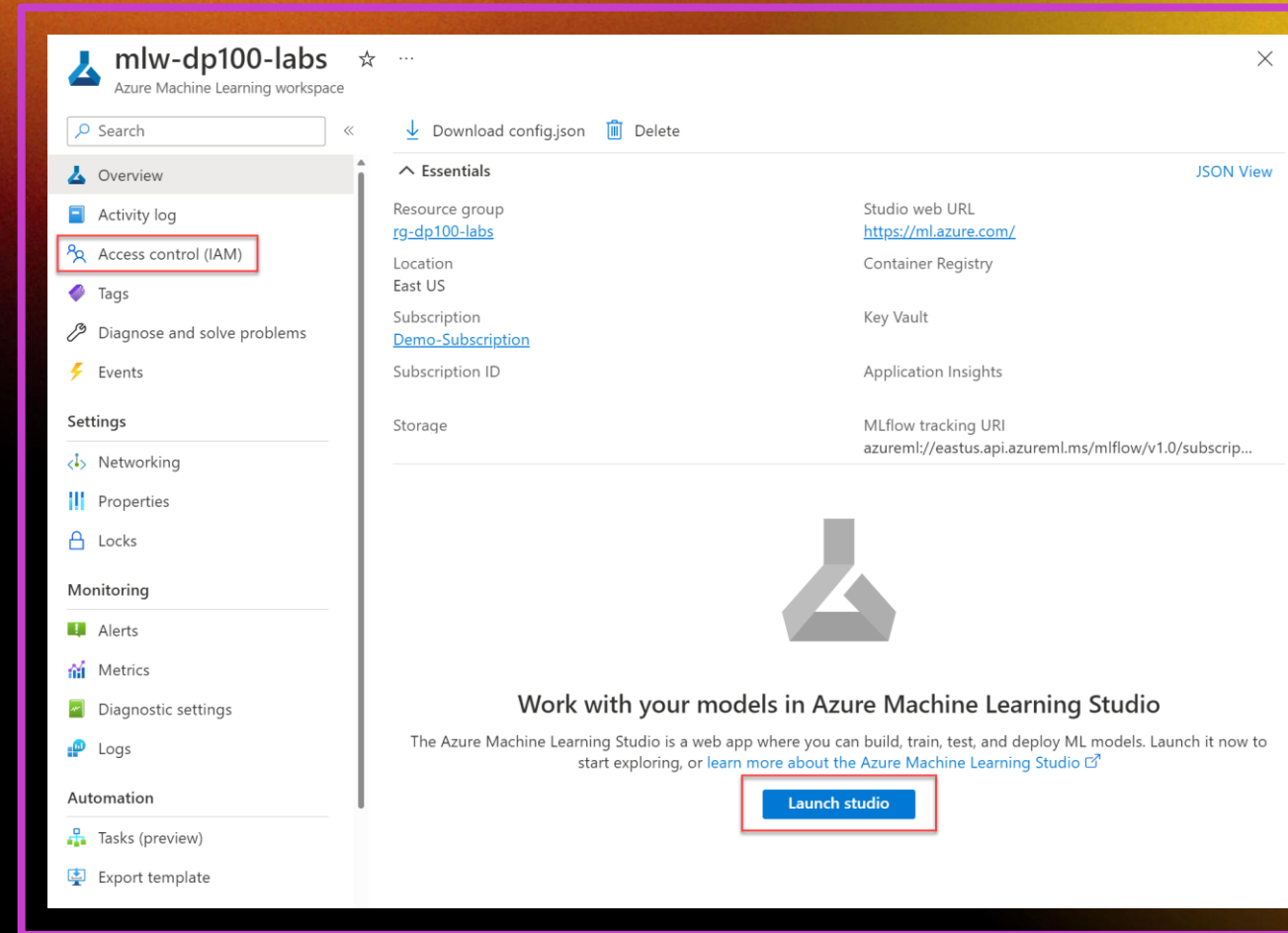


The screenshot displays the Microsoft Azure Machine Learning studio interface. The top navigation bar includes the title "Microsoft Azure Machine Learning" and icons for settings, help, and user profile. The left sidebar contains a menu with categories: "New" (Home, Author), "Assets" (Notebooks, Automated ML, Designer, Datasets, Runs, Pipelines, Models, Endpoints), and "Manage" (Compute, Datastores, Data Labeling). The main content area is titled "Azure Machine Learning studio" and features four primary action cards: "Create new" (with a plus icon), "Notebooks" (with a document icon), "Automated ML" (with a lightning bolt icon), and "Designer" (with a flowchart icon). Each card includes a brief description and a "Start now" button. Below these cards, the "My recent resources" section displays a table of recent runs.

Run	Run ID	Experiment	Status	Submitted time	Submitted by	Run type
Run 1	AutoML_6a5762d5-d3b6-...	automl_test	Completed	Oct 6, 2020 11:34 AM	Nina Baccam	Automated...

Explore the workspace In the Azure portal

- Give others access to the Azure Machine Learning workspace, using the **Access control**.
- Launch the **Azure Machine Learning studio**, an easy-to-use interface to create, manage, and use resources and assets in the workspace.



Identify Azure Machine Learning resources

The workspace

The top-level resource for Azure Machine Learning. The workspace keeps an overview of all logs, metrics, outputs, models, and snapshots of your code.

Compute resources

There are five types of compute in the Azure ML workspace: compute instances, compute clusters, Kubernetes clusters, attached computes, and serverless compute.

Datastores

All data is stored in datastores, which are references to Azure data services. Four datastores will exist by default.

Identify Azure Machine Learning assets

Models

Save trained models in the workspace. A common way to store such models is to package the model as a Python pickle file (.pkl extension).

Environments

Specify software packages, environment variables, and settings needed to run scripts. The environment is stored as an image in the Azure Container Registry when first used.

Data

You can use data assets to easily access data every time, without having to provide authentication every time you want to access it.

Components

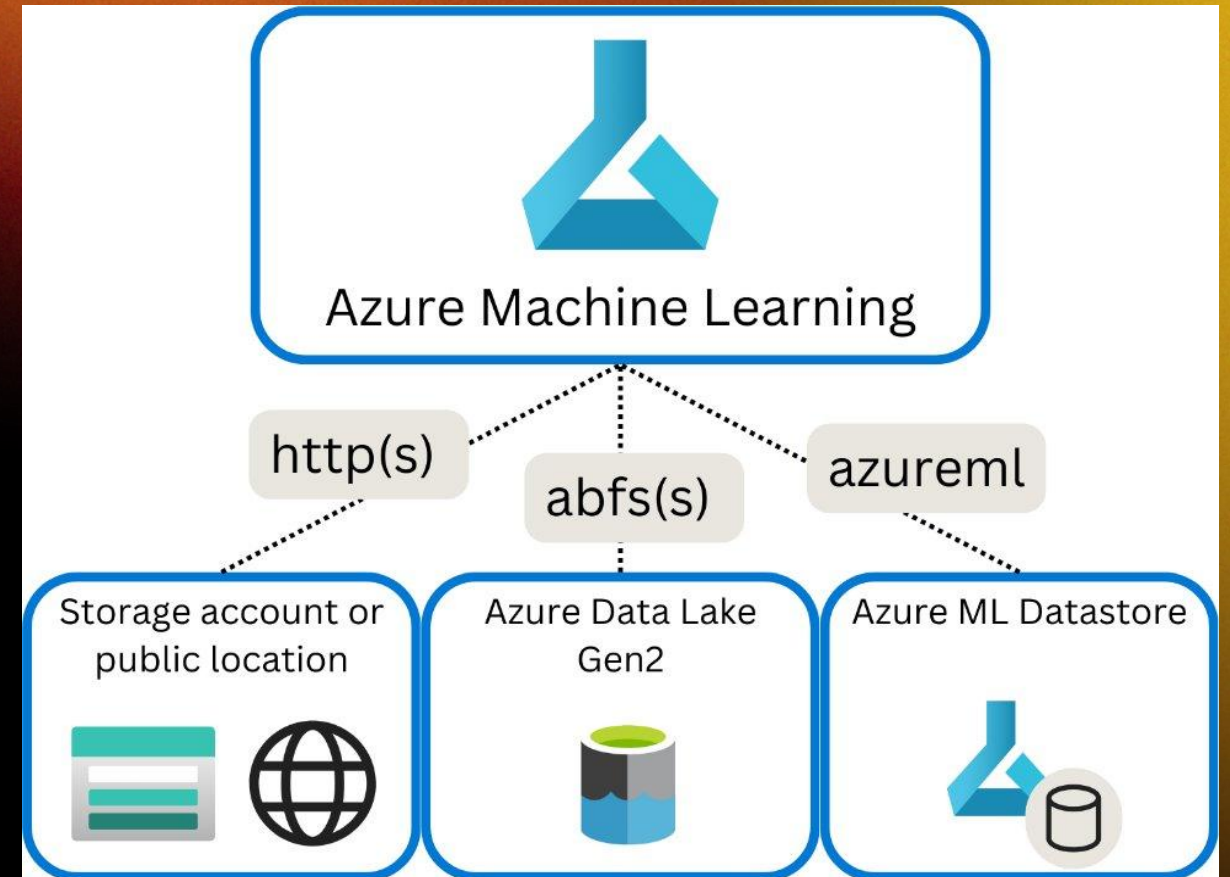
Make it easier to share code with component in a workspace.

Make data available in Azure Machine Learning

Understand URIs

A URI references the **location of your data**.

For Azure Machine Learning to connect to your data directly, you need to prefix the URI with the appropriate protocol.

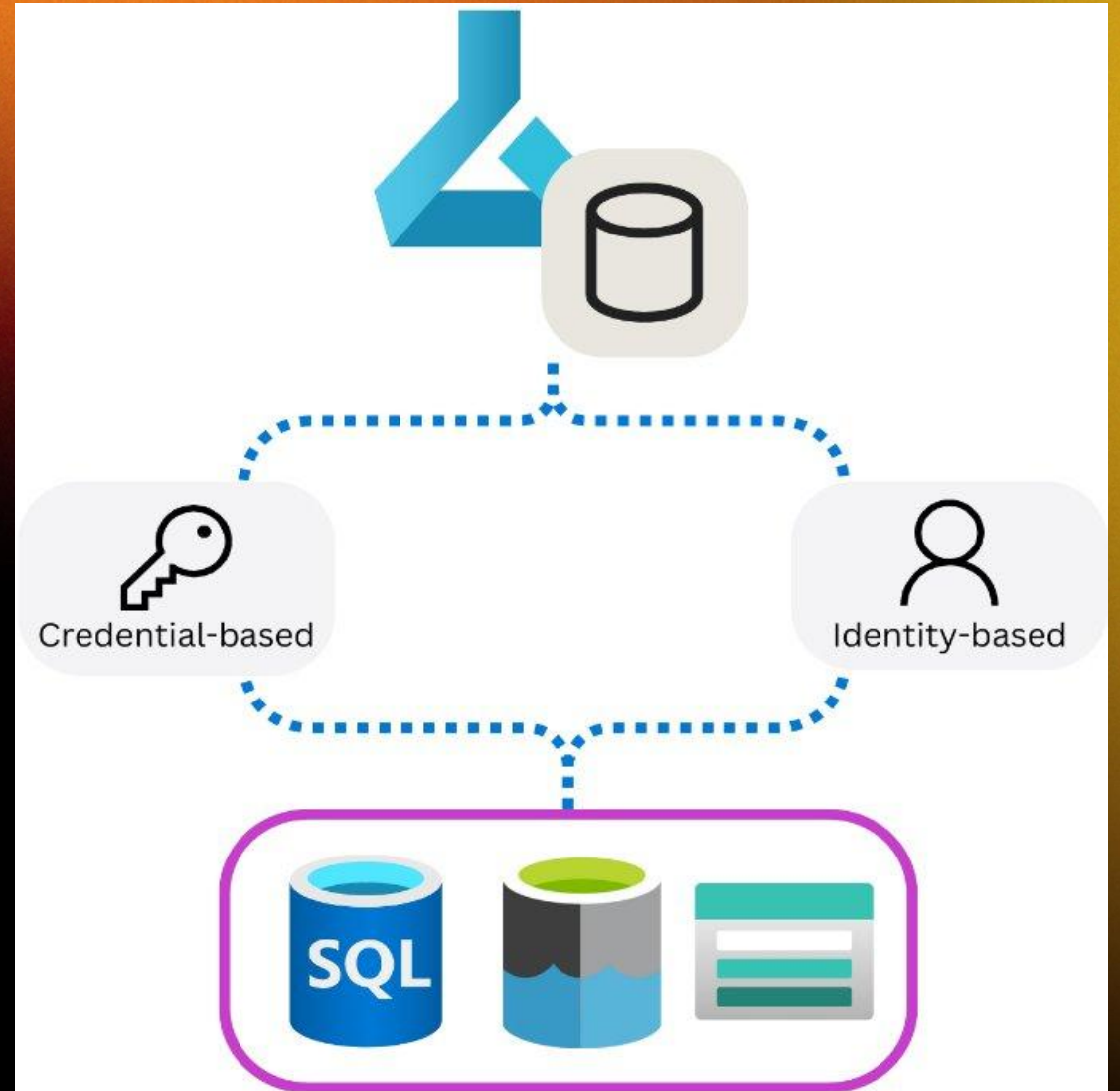


Understand datastore

Datastores are abstractions for cloud data sources, storing the connection information.

The benefits of datastores:

- Provide easy-to-use URIs to your data storage.
- Facilitates data discovery within Azure Machine Learning.
- Securely stores connection information, without exposing secrets and keys to data scientists.

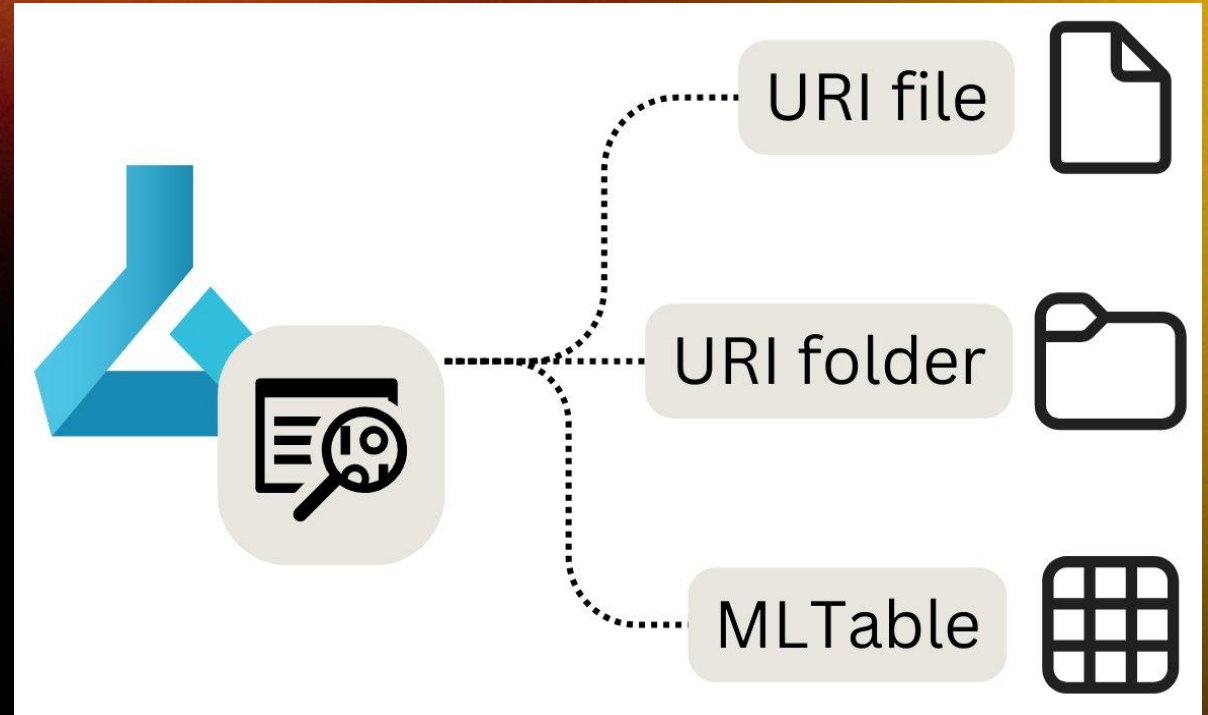


Understand data assets

Data assets are references to where the data is stored, how to get access, and any other relevant metadata.

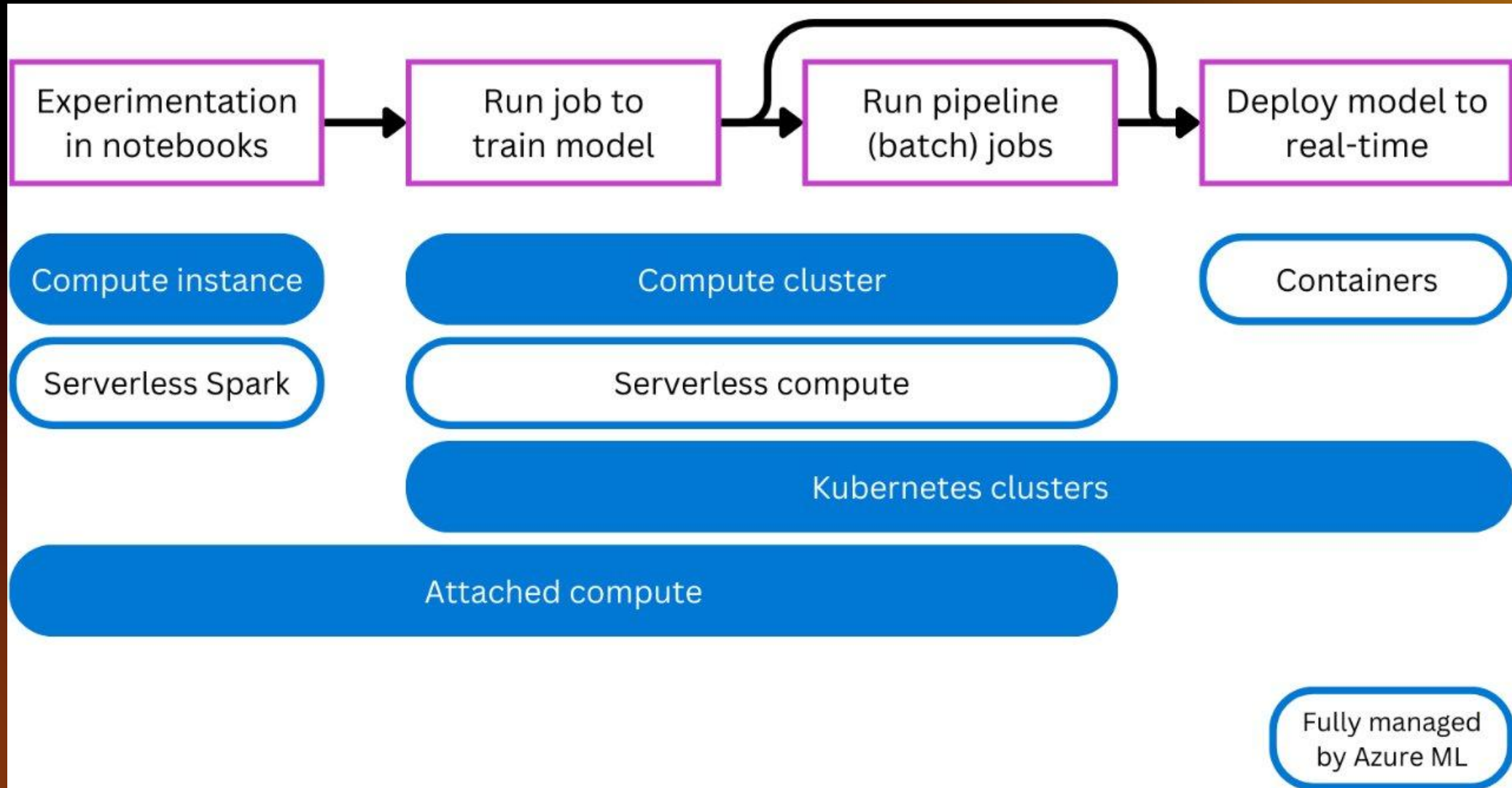
The benefits of data assets:

- Share and reuse data with other members.
- Seamlessly access data during model training (on any supported compute type) without worrying about connection strings or data paths.
- Version the metadata of the data asset.



Work with compute resources in Azure Machine Learning

Choose the appropriate compute target



Automated Machine Learning

Explore algorithms and hyperparameter values with Automated Machine Learning

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Create a new Automated ML job

✓

Select data asset

✓

Configure job

●

Select task and settings

○

Hyperparameter configuration
(Computer Vision only)

○

Validate and test

📊

Classification

To predict one of several categories in the target column. yes/no, blue, red, green.

📈

Regression

To predict continuous numeric values.

🕒

Time series forecasting

To predict values based on time.

✓

The time series forecasting method requires some additional information.

Time column * ⓘ
WeekStarting (Date) ▼

Time series identifier(s) ⓘ
☒ Autodetect
)

Frequency * ⓘ
☒ Autodetect

Forecast horizon * ⓘ
☒ Autodetect

☐ Enable deep learning ⓘ

Back

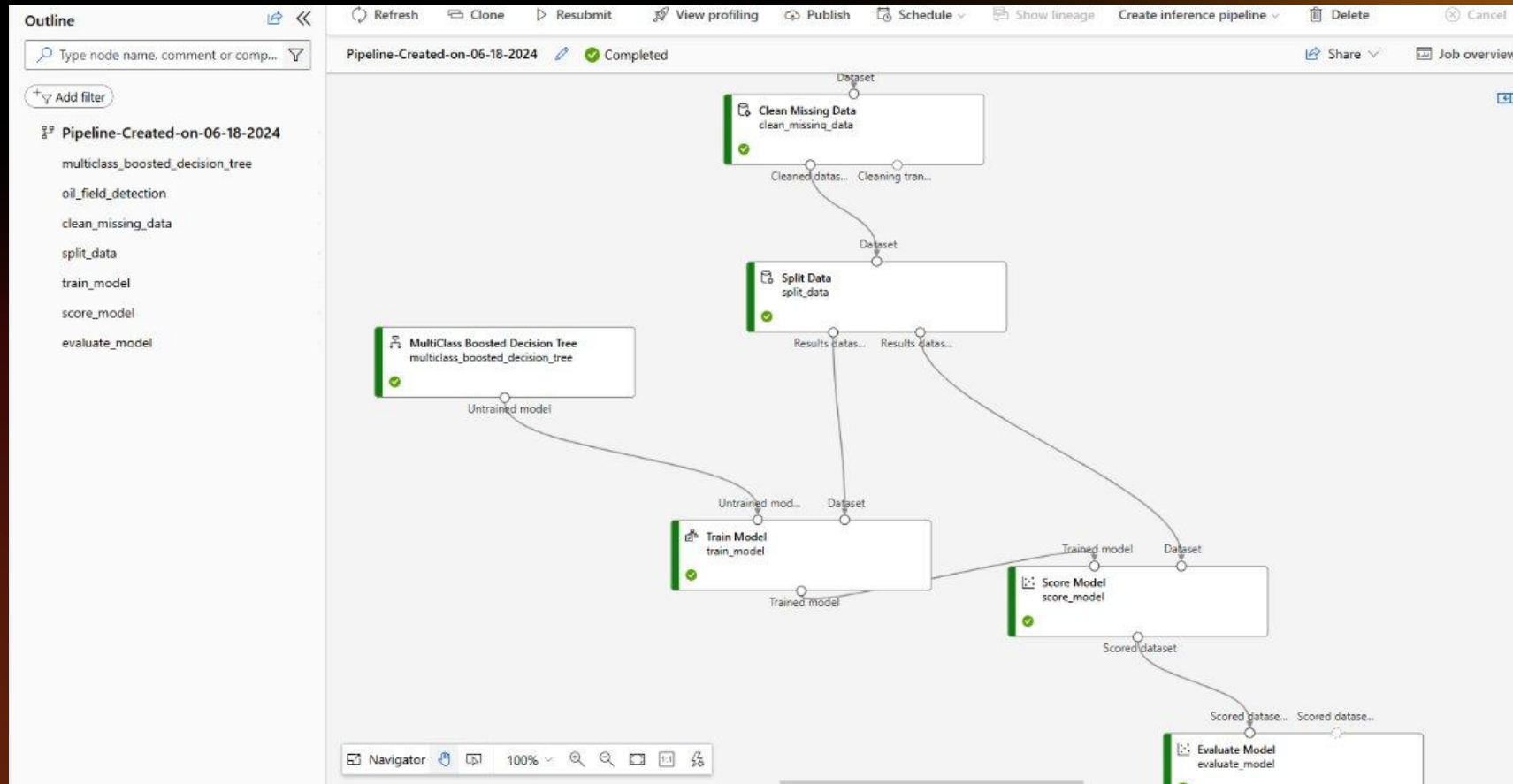
Next

Cancel

Feature Selection

Azure Machine Learning Designer

Create pipeline designer



 Search by name, tags and description

96 assets in total

Applies a mathematical operation to column values.

9/7/2020

9/7/2020

 **Apply SQL Transformation**
Microsoft
Runs a SQLite query on input datasets to transform the data.

 Microsoft

Runs a SQLite query on input datasets to transform the data.

9/7/2020

 **Clean Missing Data**
Microsoft
Specifies how to handle the values missing from a dataset.

9/7/2020

 Microsoft

Specifies how to handle the values missing from a dataset.

9/7/2020

 **Clip Values**
 Microsoft
Detects outliers and clips or replaces their values.
9/7/2020

 Microsoft

Detects outliers and clips or replaces their values.

9/7/2020

 **Convert to CSV**
Microsoft
Converts data input to a comma-separated values format.

 Microsoft

Converts data input to a comma-separated values format.

9/7/2020

 **Convert to Dataset**
 Microsoft
 Converts data input to the internal Dataset format

 Microsoft

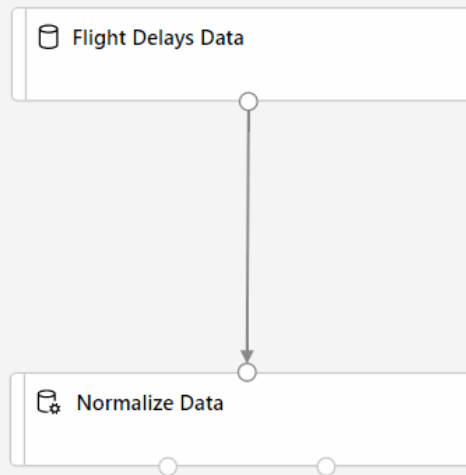
Converts data input to the internal Dataset format.

Flight Delays 

☒ Autosave on



100% ▾



Azure Machine Learning Notebook

Run a notebook

The screenshot displays the Azure Machine Learning Studio interface. On the left, a file explorer shows the directory structure: Users > madiepev > azure-ml-labs > Instructions > Labs > 01 > src. The file 'Run training script.ipynb' is selected. The main area shows the notebook content, which includes a title 'Run a training script with the Python SDK', an introduction paragraph, a 'Before you start' section with a note about installing the 'azure-ai-ml' package, a code cell with the command 'pip show azure-ai-ml', and a 'Connect to your workspace' section. The top toolbar includes buttons for running the notebook, editing in VS Code, and selecting a compute instance (currently 'ci11...').

Notebooks

Files Samples

Users

madiepev

azure-ml-labs

Instructions

Labs

01

src

Run training script.ipynb

02

03

04

05

06

07

08

09

10

Run training script.ipynb X

Edit in VS Code (pr...

Compute instance: ci11... No ...

Viewing Last saved a few seconds ago

Run a training script with the Python SDK

You can use the Python SDK for Azure Machine Learning to submit scripts as jobs. By using jobs, you can easily keep track of the input parameters and outputs when training a machine learning model.

Before you start

You'll need the latest version of the **azureml-ai-ml** package to run the code in this notebook. Run the cell below to verify that it is installed.

Note: If the **azure-ai-ml** package is not installed, run `pip install azure-ai-ml` to install it.

+ Code + Markdown

```
1 pip show azure-ai-ml
```

Press shift + enter to run

Connect to your workspace

With the required SDK packages installed, now you're ready to connect to your workspace.

To connect to a workspace, we need identifier parameters - a subscription ID, resource group name, and workspace name. The resource group name and workspace name are already filled in for you. You only need the subscription ID to complete the command.

To find the necessary parameters, click on the subscription and workspace name at the top right of the Studio. A pane will open on the right.

Microsoft Azure Machine Learning

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documentationteam
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*my-notebook.ipynk

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Editors

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Compute: my-cpu-cluster - Running

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Python 3.6 - AzureML

my-cpu-cluster · Jupyter kernel idlePython 3.6.9

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Environment Setup: **Azure Machine Learning**

Compute

Prerequisites: Make sure your account already has access and can sign in to the Azure Machine Learning workspace

Navigate to Compute Instance and Compute Cluster

Create Compute Instance/Cluster

Set configuration compute

Kernel, ML version

Prerequisites: Make sure to having compute instance and some notebook

Start compute and open a notebook

Choose kernel based on use case

Exercise: Azure Machine Learning Hands On

Use Case

Geothermal Gradient Prediction

Objective

This Project is aimed at predicting geothermal characteristics for Colombia, specifically the geothermal gradient, based on available geological and geophysical data. We employ machine learning techniques to make these predictions. This project focuses on predicting the Apparent Geothermal Gradient ($^{\circ}\text{C}/\text{Km}$) as an essential factor in geothermal exploration.

Methodology

The project utilizes geospatial data, geophysical information, and geothermal measurements. These datasets are located in the data folder of this repository. The data includes information on well depths, temperatures, geological features, and proximity to volcanic structures.

Use Case

Geothermal Gradient Prediction

THE MACHINE LEARNING CANVAS (V1.1)				
Designed for: _____		Designed by: _____		Date: _____ Iteration: _____
PREDICTION TASK  Entity on which predictions are made? Possible outcomes? Wait time before observation? Predict the geothermal characteristics especially in gradient based on geophysical and geological data.	DECISIONS  How are predictions turned into proposed value for the end-user? Mention parameters of the process / application that does that. Geothermal is one of essential factors in geothermal exploration.	VALUE PROPOSITION  Who is the end-user? What are their objectives? How will they benefit from the ML system? Mention workflow/interfaces. The end user is the geothermal industry actors (geologist, geophysics, etc.)	DATA COLLECTION  Strategy for initial train set & continuous update. Mention collection rate, holdout on production entities, cost/constraints to observe outcomes. The data is included information of geological, geophysical (well depths, temperatures, proximity to volcanic structures)	DATA SOURCES  Where can we get (raw) information on entities and observed outcomes? Mention database tables, API methods, websites to scrape, etc. Dataset can be obtained from Geothermal Power Database (International Geothermal Association)
IMPACT SIMULATION  Can models be deployed? Which test data to assess performance? Cost/gain values for (in)correct predictions? Fairness constraint? Model can be deployed as restapi, azure mlops. So it can be used for further application (web/mobile app)	MAKING PREDICTIONS  When do we make real-time / batch pred.? Time available for this + featurization + post-processing? Compute target? We make the batch prediction since it's not time series data. And the compute target can be compute instance, compute cluster or kubernetes	BUILDING MODELS  How many prod models are needed? When would we update? Time available for this (including featurization and analysis)? we can use any regression models as long as the model give an accurate result.		
MONITORING  Metrics to quantify value creation and measure the ML system's impact in production (on end-users and business)? Models can be monitored using mlflow.				

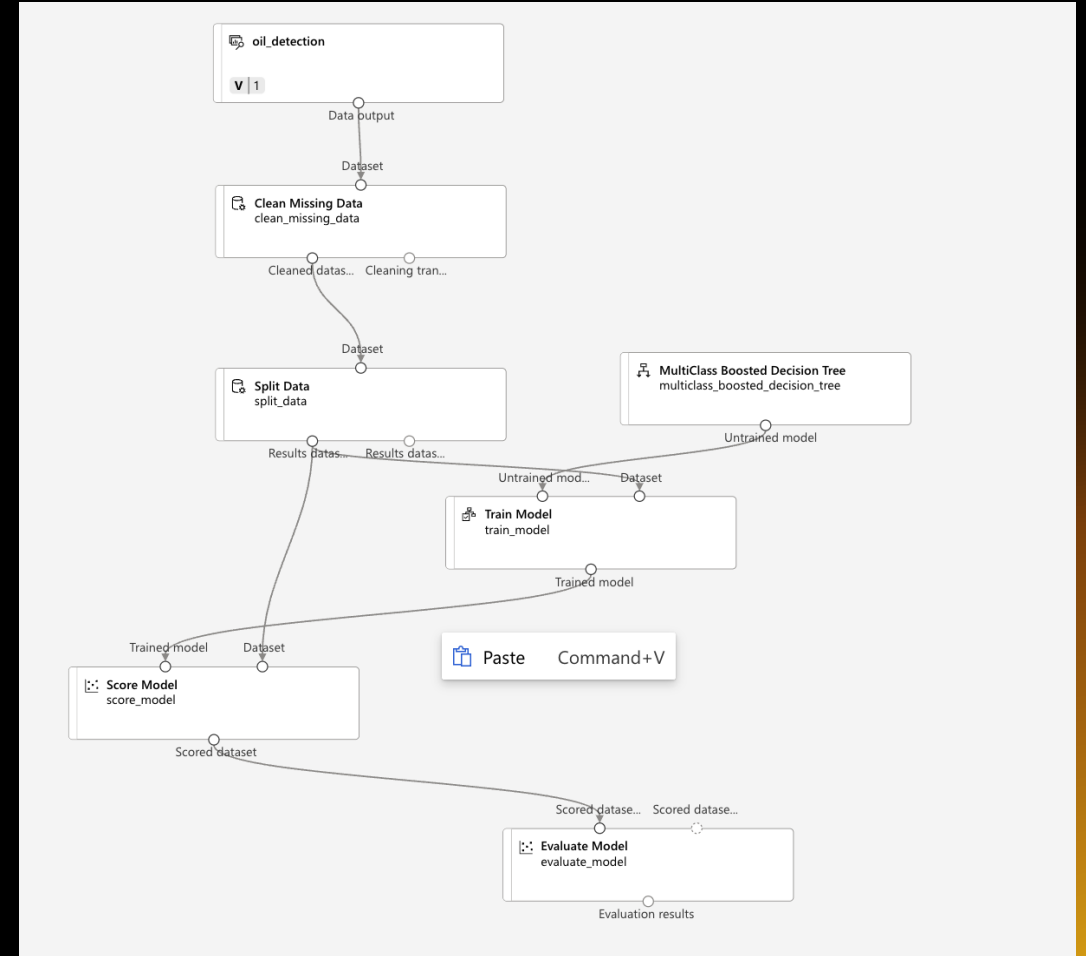
Use Case

Geothermal Gradient Prediction using AML Designer

Technically the steps we do in Azure Machine Learning Designer resembles the steps on Azure Machine Learning Notebooks / python file. But we transform it to be low code mode, using some flow boxes.

Methods :

- Select AML Designer menu
- Create new pipeline/designer projects
- Follow the steps as shown on figure



Use Case

Geothermal Gradient Prediction using Auto ML

Method :

- Open Azure Automated ML
- Create new project or pick existing project and pick ML type (classification, regression, etc)
- Connect to your dataset.
- Train your model.

Authoring

- 📄 Notebooks
- ⚡ Automated ML
- 🏗 Designer
- >_ Prompt flow
- 🔍 Tracing

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Automated ML

Let Automated ML train and find the best model based on your data without writing a single line of code. [Learn more about Automated ML](#)

+ New Automated ML job Refresh

Recent Automated ML jobs [View all experiments](#)

Display name	☆ Experiment	Status	Created on	Duration	Created by	Compute targ...	Tags
preprocessed-text	first-experiment	✔ Completed	Aug 15, 2024 1:16 PM	6h 12m 26s	Muhammad F...	Serverless	dynamic_allowlist ...
raw-sample-	first-experiment	✖ Failed ⓘ	Aug 15, 2024 11:43 AM	12s	Muhammad F...	Serverless	_azureml.Compute ...
kaifinance-usecase1-demo-tdpclassifica	first-experiment	✖ Canceled	Aug 15, 2024 10:03 AM	3h 13m 37s	Muhammad F...	Serverless	dynamic_allowlist ...

Authoring

- 📄 Notebooks
- ⚡ Automated ML
- 🏗 Designer
- >_ Prompt flow
- 🔍 Tracing

1

2

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PT. Nawa Darsana Teknologi > aml-development > Training job

Submit an Automated ML job

- Training method
- Basic settings
- Task type & data
- Task settings
- Compute
- Review

Basic settings

Let's start with some basic information about your training job.

Job name *

Experiment name *

Select existing ☒ Create new ☐

Existing experiment *

Description

Tags

Name Value Add

Back Next Cancel

3

PT. Nawa Darsana Teknologi > aml-development > Training job

Submit an Automated ML job

- Training method
- Basic settings
- Task type & data
- Task settings
- Compute
- Review

Task type & data

Choose the type of task that you would like your model to

Select task type * ⓘ

Select task type

Classification

To predict one of several categories yes/no, blue, red, green.

Regression

To predict continuous numeric values.

Time series forecasting

To predict values based on time.

Natural language processing

To predict based on text-only data types using multi-class or multi-label classification or named entity recognition.

Computer vision

To predict using multi-class or multi-label image classification, object detection, and instance segmentation.

4

Thank You